Lab Assignment 19.2 – Code Translation

Annapu Vasantha Shoba Rani

2303A51395

Task 1: Python to Java Conversion (File Handling)

Question

Convert a Python file handling program into Java.

Prompt

"Translate a Python file read/write program into Java with proper file handling and exception handling."

Code

Python Code

```
with open("sample.txt", "w") as f:
    f.write("Hello World")

with open("sample.txt", "r") as f:
    print(f.read())
```

Java Code

```
import java.io.*;

public class FileExample {
    public static void main(String[] args) {
        try {
            FileWriter writer = new FileWriter("sample.txt");
            writer.write("Hello World");
            writer.close();

            BufferedReader reader = new BufferedReader(new FileReader("sample.txt"));
            String line = reader.readLine();
```

```
            System.out.println(line);
            reader.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

Output

Hello World

Code Explanation

- Python file handling uses with open()
- Java uses FileWriter and BufferedReader
- Exception handling required in Java
- Both produce same output

Task 2: C++ to Python Conversion (Sorting)

Question

Convert C++ sorting program into Python.

Prompt

"Translate C++ bubble sort program into Python and verify correctness."

Code

C++ Code

```
#include <iostream>
using namespace std;

int main() {
```

```
    int arr[4] = {5, 2, 9, 1};
    for(int i=0;i<4;i++){
        for(int j=0;j<4-i-1;j++){
            if(arr[j] > arr[j+1]){
                swap(arr[j], arr[j+1]);
            }
        }
    }
    for(int i=0;i<4;i++)
        cout<<arr[i]<<" ";
}
```

Python Code

```
arr = [5, 2, 9, 1]

for i in range(len(arr)):
    for j in range(len(arr)-i-1):
        if arr[j] > arr[j+1]:
            arr[j], arr[j+1] = arr[j+1], arr[j]

print(arr)
```

Output

[1, 2, 5, 9]

Code Explanation

- Same bubble sort logic used
- Python uses list swapping directly
- C++ uses swap() function
- Output remains identical

Task 3: SQL to Pandas (GROUP BY)

Question

Convert SQL aggregation query into Pandas.

Prompt

"Translate SQL GROUP BY query into Pandas groupby() operation."

Code

SQL Query

```
SELECT department, SUM(sales)
FROM employees
GROUP BY department;
```

Python Code

```
import pandas as pd

data = {
    "department": ["HR", "IT", "HR", "IT"],
    "sales": [100, 200, 150, 300]
}

df = pd.DataFrame(data)

result = df.groupby("department")["sales"].sum()
print(result)
```

Output

```
HR    250
IT    500
```

Code Explanation

- SQL GROUP BY → Pandas groupby()

- SUM aggregation applied
- Same grouped result achieved

Task 4: Multi-Language Snippet Converter

Question

Convert Python API into another language.

Prompt

"Convert Python Flask API into Node.js Express API."

Code

Python (Flask)

```
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "Hello API"

app.run()
```

Node.js (Express)

```
const express = require('express');
const app = express();

app.get('/', (req, res) => {
    res.send("Hello API");
});

app.listen(3000);
```

Output

Hello API

Code Explanation

- Flask routes → Express routes
- Python functions → JavaScript callbacks
- Same API functionality

Task 5: Python to C++ (Even/Odd)

Question

Convert Python function to check even/odd into C++.

Prompt

"Translate Python even/odd function into C++ with input handling."

Code

Python Code

```
def check_even_odd(n):
    if n % 2 == 0:
        return "Even"
    else:
        return "Odd"

print(check_even_odd(4))
```

C++ Code

```
#include <iostream>
using namespace std;

int main() {
    int n;
```

```
    cin >> n;

    if(n % 2 == 0)
        cout << "Even";
    else
        cout << "Odd";

    return 0;
}
```

Output

Input: 4

Output: Even

Code Explanation

- Modulus operator checks divisibility
- Python uses function, C++ uses main
- Same logic implemented
- Works for multiple inputs